JAN-G-439

31 DECEMBER 1946

JOINT ARMY-NAVY SPECIFICATION

GILDING METAL (95/5 BRASS); SHEET AND STRIP

This specification was approved by the War Department and the Navy Department for use of procurement services of the Army and the Navy and supersedes the following specifications:

U. S. Army	Navy Department
57–171–2	Bureau of Ordnance OS 819
12 Dec. 1939	18 June 1941

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATION

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material, together with Appendix II, Metals, Part A, Definitions and Tests.[1]

FEDERAL SPECIFICATION

QQ–M–151—Metals; General Specification for Inspection Of.

A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[1]

B. GRADE, FORMS, AND CONDITION.

B–1. *Grade and forms.*—Gilding metal shall be furnished in one grade and in the following forms, as specified (see par. H–2):

Sheet.
Strip.

B–2. *Condition.*—Gilding metal shall be furnished in the following condition:

Drawing anneal.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—Gilding metal shall be made from high grade copper and high grade zinc or from high grade ingot metals and clean scrap of known composition, which will produce material meeting the requirements of this specification.

C–2. *Workmanship.*—

C–2a. The metal shall be free from laminations, scale, dented or bent edges, pipes, slivers, laps, cracks, deep scratches, wrinkles, and other defects which will render the metal unsuitable for the purpose intended.

[1] Applicable only to Navy purchases.

Source: https://assist.dla.mil -- Downloade
Check the source to verify that this is the cu

C–2b. The metal shall be uniform in thickness, composition, and anneal and shall be free from kinks and warps. Coiled material shall be loose enough to feed satisfactorily into the cupping machine.

C–2c. Sheet and strip, when ordered for spinning or drawing, shall be of such quality as to present, after spinning or drawing, a good smooth surface that will not be spotted or open-grained.

C–2d. Sheet and strip, when ordered for either shallow or deep cupping, shall not have directional properties which result in the formation of ears when the material is cupped.

C–2e. Workmanship shall be first-class in every respect.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Chemical composition.*—

E–1a. Gilding metal shall meet the requirements shown in table I.

TABLE I.—*Chemical requirements*

Percent composition				
Copper	Lead (maximum)	Iron (maximum)	Total elements other than copper or zinc (maximum)	Zinc
94.0 — 96.0	0.03	0.05	0.13	Remainder

E–1b. An analysis of each lot shall be furnished by the contractor, showing the percentage of each element designated in table I.

E–2. *Grain size.*—Gilding metal shall meet the grain size requirements shown in table II.

TABLE II.—*Grain size requirements*

Grain size, millimeter		
Nominal	Minimum	Maximum
.035	.025	.050

E–3. *Dimensions and permissible variations.*—

E–3a. The material shall be furnished in flat lengths or coils, as specified (see par. H–2). The permissible variation at any point in thickness of sheet or strip to be used for general purposes, shall be as shown in table III.

Source: https://assist.dla.mil --
Check the source to verify that thi

E–3b. For the manufacture of components for military ammunition, where departures from table III are necessary, dimensions shall be as specified in the contract or order (see par. H–2).

TABLE III.—*Permissible variations in thickness*

Thickness, inch	Permissible variations, inch, plus and minus						
	8 inch and under in width	Over 8 to 14 inch inclusive in width	Over 14 to 20 inch inclusive in width	Over 20 to 28 inch inclusive in width	Over 28 to 36 inch inclusive in width	Over 36 to 48 inch inclusive in width	Over 48 to 60 inch inclusive in width
0.004 inch and under	0.0003	0.0006	0.0013	.....	.....	.....	.....
Over 0.004 to 0.006 incl.	0.0004	0.0008	0.0013	.....	.....	.....	.....
Over 0.006 to 0.009 incl.	0.0006	0.0010	0.0015	.....	.....	.....	.....
Over 0.009 to 0.013 incl.	0.0008	0.0013	0.0018	0.0025	0.003	0.0035	0.004
Over 0.013 to 0.017 incl.	0.0010	0.0015	0.002	0.0025	0.003	0.0035	0.0045
Over 0.017 to 0.021 incl.	0.0013	0.0018	0.002	0.003	0.0035	0.004	0.005
Over 0.021 to 0.026 incl.	0.0015	0.002	0.0025	0.003	0.0035	0.004	0.005
Over 0.026 to 0.037 incl.	0.002	0.002	0.0025	0.0035	0.004	0.005	0.006
Over 0.037 to 0.050 incl.	0.002	0.0025	0.003	0.004	0.005	0.006	0.007
Over 0.050 to 0.073 incl.	0.0025	0.003	0.0035	0.005	0.006	0.007	0.008
Over 0.073 to 0.130 incl.	0.003	0.0035	0.004	0.006	0.007	0.008	0.010
Over 0.130 to 0.187 incl.	0.0035	0.004	0.0045	0.007	0.008	0.010	0.012

Source: https://assist.dla.mil -- Downloade
Check the source to verify that this is the cu

E–3c. The width of slit sheet and strip and slit metal with rolled edges shall not vary from the ordered width by more than the amount shown in table IV.

TABLE IV.—*Permissible variations in width*

Width, inches	Permissible variations, inch, plus and minus	
	0.004 to 0.032 inch incl. in thickness	Over 0.032 to 0.187 inch incl. in thickness
2 inches and under	0.005	0.010
Over 2 to 8, inclusive	0.008	0.013
Over 8 to 14, inclusive	0.010	0.015
Over 14 to 20, inclusive	0.013	0.018

E–3d. The width of square-sheared metal shall not vary from the ordered width by more than the amount shown in table V.

TABLE V.—*Permissible variations in width*

(Square-sheared metal up to 120 inches incl. in length.)

Width, inches	Permissible variations, inch, plus and minus		
	1/16 inch and under in thickness	Over 1/16 to 1/8 inch incl. in thickness	Over 1/8 inch in thickness
20 inches and under	1/32	3/64	1/16
Over 20 to 36 incl.	3/64	3/64	1/16
Over 36 to 60 incl.	1/16	1/16	1/16

E-3e. Permissible variations in width of sawed metal in all gages shall be + 0.032 and — 0.000 inch.

E–3f. Permissible variations in width of rolled-edge metal not previously slit shall be as shown in table VI.

TABLE VI.—*Permissible variations in width*

(Rolled-edge metal.)

Width, inches	Permissible variations, inch, plus and minus		
	0.032 inch and under in thickness	Over 0.032 to 0.125 inch incl. in thickness	Over 0.125 inch in thickness
⅛ inch and under	0.002	0.002	....
Over ⅛ to ⅜, incl.	0.003	0.003	0.004
Over ⅜ to 1, incl.	0.004	0.004	0.006
Over 1 to 2, incl.	0.005	0.007	0.010
Over 2 to 4, incl.	....	0.015	0.015
Over 4 to 6, incl.	....	0.020	0.020
Over 6	....	0.025	0.025

Source: https://assist.dla.mil -- Check the source to verify that th

E–4. *Straightness.*—

E–4a. *Flat lengths.*—The edgewise curvature (depth of arc) in any 72-inch length of sheet or strip shipped flat shall not exceed 0.5 inch.

E–4b. *Coils.*—The edgewise curvature (depth of arc) in any 72 inch length of coil shall not exceed 1 inch.

E–5. *Length.*—

E–5a. *Exact lengths.*—Sheet and strip may be ordered to exact lengths in all sizes, in which case a variation of plus 0.250 inch in length will be permitted.

E–5b. *Stock lengths.*—When sheet and strip 12 inches or less wide are ordered in stock lengths, the pieces shipped shall be cut to the nominal or stock length plus or minus one-half inch as a maximum. The shortest acceptable lengths, the maximum permissible percentage by weight of short lengths, and the required minimum percentage by weight of stock lengths in any one shipment shall be as shown in table VII.

TABLE VII.—*Permissible variations in length*

Nominal or stock length ordered (feet)	Required minimum percentage, by weight, of stock lengths	Maximum permissible percentage, by weight, of short lengths				
		In lengths 10 feet and over	In 8- to 10-feet lengths	In 6- to 8-feet lengths	In 4- to 6-feet lengths	In 2- to 4-feet lengths
10 to 12	60	40	30	20	10	....
10	60	....	40	30	20	10
8	70	....	....	30	20	10
6	80	....	....	....	20	10

E–6. *Finished (rolled) edges.*—

E–6a. Where finished (rolled) edges are specified in the contract or order (see par. H–2) the following descriptions shall apply:

"Square edge" material shall have finished edges with sharp, square corners, without bevel or rounding of any sort.

"Round edge" material shall have finished edges, semicircular in form, the diameter of the circle forming the edge being equal to the thickness of the material.

"Round corner" material shall have square finished edges, with corners rounded to a definite radius which, unless otherwise specified in the contract or order, may be any radius less than one half the thickness of the material.

E–6b. Where no description of any required form of edge is given, edges produced by slitting, sawing, or shearing will be acceptable.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Inspection.*—Inspection and tests may be made at the place of manufacture or at the point of delivery, as specified in the contract or order (see par. H–2).

Source: https://assist.dla.mil -- Downloade
Check the source to verify that this is the cu

F–2. *Lot.*—Unless otherwise specified, a lot shall consist of not more than 5000 pounds of material of the same form and size, submitted for inspection at the same time.

F–3. *Selection of samples.*—At least 10 flat lengths or coils shall be selected by the inspector to represent each lot.

F–4. *Gaging.*—Each flat length or coil, selected in accordance with paragraph F–3, shall be gaged for dimensions. Measurements for thickness shall be made over the entire width of the flat length or coil, and the thickness shall be measured in at least three different places, not less that 1 foot from each end, and at such other points as may be selected by the inspector.

F–5. *Sample for Government analysis.*—

F–5a. From each flat length or coil, selected in accordance with paragraph F–3, the inspector shall obtain an equal amount of metal which shall be thoroughly mixed, and not less than 2 ounces of the resultant mixture shall be used for chemical analysis.

F–5b. Drillings, millings, or sawings shall be taken on material 0.040 inch or more in thickness.

F–5c. The drill, cutter, saw, or other tool used for sampling the metal shall be thoroughly cleaned. No lubricant shall be used in the operation, and the drillings, millings, sawings, or clippings shall be carefully treated with a magnet to remove any particles of iron introduced in taking the sample.

F–6. *Government analysis.*—

F–6a. At the option of the purchaser, analysis for determining elements other than copper and zinc may be made by the spectrographic method, provided that in case of dispute, analysis by standard gravimetric or volumetric methods shall be the basis for acceptance.

F–6b. Analysis shall regularly be made only for copper, lead, and iron. If, however, the presence of elements other than copper, lead, iron, and zinc is suspected or indicated in the course of routine analysis, further analysis shall be made to determine that the total of these other elements is not in excess of the limits specified.

F–7. *Grain size determination.*—

F–7a. Two samples for grain size measurements shall be selected from each lot. Unless otherwise specified, the average grain size shall be determined on a plane parallel to the surface of the flat lengths or coils.

F–7b. The grain size shall be determined by direct comparison with standard micrographs furnished by the purchaser or by procedure outlined in paragraph F–7c. In case of dispute, the diameter of the average grain shall be determined as prescribed in paragraph F–7c. In counting individual grains, the original crystal, including twinned layers, shall constitute one grain.

F–7c. *Method of grain size measurement.*—

F–7c(1). A circle 79.8 mm. in diameter (or a rectangle of the same area, 5,000 sq. mm.) shall be drawn on the rough side of a ground-glass screen, the center of the circle being near the center of the rectangular section of the screen. This ground glass shall

Source: https://assist.dla.mil -
Check the source to verify that t

be mounted in a frame which fits a metallographic camera, the smooth side of the glass being on the outside. When the image of the specimen for grain size determination is focused on the screen, the circle will be plainly visible and its circumference shall be well within the image.

F–7c(2). When the image is properly focused the grains intersected by the circumference of the circle (or periphery of the rectangle) shall be checked and counted. Since the check marks must be made on the smooth side of the glass, a soft red pencil, such as used in laboratories for marking beakers and flasks, will be found satisfactory. The marks used to indicate the boundary grains are usually short, straight lines intersecting the circumference of the circle and perpendicular to it. The completely included grains shall be next checked and counted, after which the red marks may be erased from the glass with a dry cloth. The specimen may then be moved and other measurements made as desired.

F–7c(3). One-half the number of grains intersected by circumference of the circle (or periphery of the rectangle), added to the number of completely included grains, gives the number of equivalent whole grains within the circle.

F–7c(4). *Formulas.*—The following formulas shall be used in computing the grain size:

$$x = \tfrac{1}{2} w + z$$

$$f = \frac{m^2}{5{,}000}$$

$$n = fx$$

$$d = \frac{1}{\sqrt{n}}$$

where

w = number of boundary grains
z = number of completely included grains
x = equivalent number of whole grains in 5,000 square millimeters
m = magnification used
f = multiplier to obtain number of grains per square millimeter
n = number of grains per square millimeter
d = diameter of average grain in millimeters.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packing.*—

G–1a. All material shall be properly segregated as to size when loaded for shipment.

G–1b. Flat lengths and coils shall be packed in boxes or crates so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery.

G–2. *Marking.*—Each shipping container shall be marked with the name of the material, form, size, and the weight contained therein, as defined by the contract or order under which ship-

Source: https://assist.dla.mil -- Downloade
Check the source to verify that this is the cu

ment is made, the name of the contractor, and the number of the contract or order. In addition, shipments for the Navy shall be marked in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—The material covered by this specification is used in the manufacture of primer and detonator cups for artillery ammunition and bombs.

H–2. Requests, requisitions, schedules, and contracts or orders should cover the following features:

(a) Title, number, and date of the specification.
(b) Form and dimensions of gilding metal required (see pars. B–1 and E–3).
(c) Whether cut flat lengths or coils are desired (see par. E–3a).
(d) Special tolerances if required (see par. E–3b).
(e) Whether finished edges are desired (see par. E–6).
(f) Place of inspection and tests (see par. F–1).

H–3. Copies of Joint Army-Navy specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications Used by the War Department." Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Agencies within the War Department will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established War Department channels. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–4. Copies of Joint Army-Navy specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Depot, Bayonne, N. J. Both the title and identifying symbol number should be stipulated when requesting copies of specifications.

H–5. Copies of this Joint Army-Navy specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, War Department, Washington 25, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

ARMY: O
NAVY: OASbS.

[JAN–G–439]

Source: https://assist.dla.mil
Check the source to verify that